

Lab B — Microphone scattering

Reference version (for checking)

34870 Electroacoustics

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Reference version, not the hand-in. This is a complete worked version of the report, meant to be used for checking the group's own report (`LabB_report.tex`): the numbers, the figures and the reasoning. Every number comes from the processed data (`LabB/data/process_labB_output.txt`, `LabB/bem/bem_results.csv`) or is marked as an assumption. The group writes its own text.

1 Purpose and theoretical basis

The experiment investigates how a microphone-shaped rigid body changes the pressure at its front face. A wooden, enlarged microphone model makes this scattering measurable using a small microphone. The angle-dependent response is compared with a boundary element method (BEM) calculation and with microphone handbook free-field corrections.

For a body radius a , the dimensionless frequency is

$$ka = \frac{2\pi fa}{c}, \quad a = \frac{D_{\text{mock}}}{2}. \quad (1)$$

Here c is the sound speed. Similar geometries are compared at equal ka . The measured voltage ratio and the normalized scattering response are

$$H_{\theta}(f) = \frac{V_{\text{mic},\theta}(f)}{V_{\text{ref},\theta}(f)}, \quad H_0^{\text{ref}}(f) = \frac{V_{\text{mic,no mock}}(f)}{V_{\text{ref,no mock}}(f)}, \quad (2)$$

$$H_{\text{norm},\theta}(f) = \frac{H_{\theta}(f)}{H_0^{\text{ref}}(f)}, \quad \Delta L_{\theta}(f) = 20 \log_{10} |H_{\text{norm},\theta}(f)|. \quad (3)$$

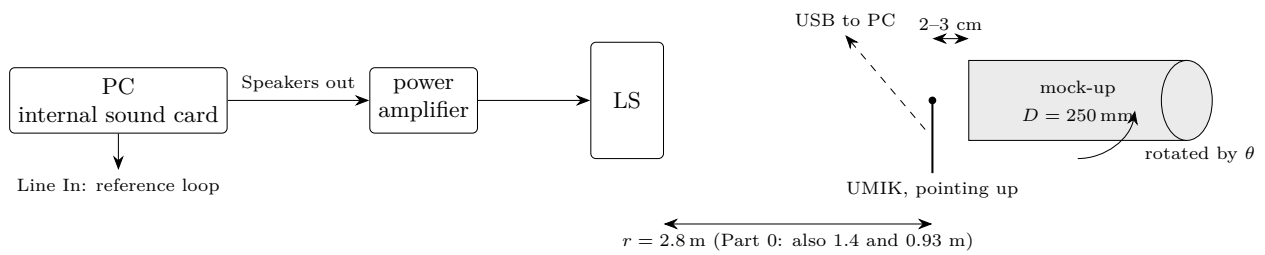
The subscript 0 in H_0^{ref} denotes the reference run, not zero-degree incidence. Normalization removes the common measurement-chain response only when its settings and the loudspeaker and UMIK positions remain unchanged.

For $ka \ll 1$ the body is small compared with the wavelength, the wave flows around it and $\Delta L_{\theta} \rightarrow 0$ at every angle. As ka approaches 1 the front face starts to reflect; head-on the reflected wave doubles the pressure (6 dB for an infinite rigid wall) and the wave diffracted at the rim of the face reaches the centre from the whole edge with the same delay, so the centre-of-face pressure exceeds 6 dB and reaches about 10 dB near $ka \approx 3$. At grazing incidence (90°) the face does not obstruct the wave and the increase stays within about 2 dB. The loudspeaker radiates a spherical wave; over the body it approximates a plane wave when the source distance r is large compared with the body length and when the level follows $1/r$ (Part 0). With $r = 2.8$ m and a body of 0.855 m, the incident phase along the body is not exactly planar; the effect is small at the low ka where the comparison is made.

2 Equipment and geometry

Item	Recorded identity or setting
PC / internal sound card	Lab desktop, internal card (Line In reference, Speakers output); course routine <code>meas_mag_spec2_SoundCard_LabB</code> called from <code>labB.m</code>
UMIK MiniDSP / USB microphone	S/N 708-0332, correction for 90° incidence on the UMIK (it points up)
Loudspeaker / amplifier	Chamber loudspeaker; amplifier setting not noted, unchanged for the whole session
Wooden microphone mock-up	$D_{\text{mock}} = 250$ mm, length 855 mm — nominal values from the brief, not measured with the tape
Mock-up back end	Round
Loudspeaker–UMIK distance	$r = 2.8$ m from the loudspeaker front, confirmed by the $1/r$ check; driver alignment and height not noted
UMIK tip–front-face gap	Approximately 2 cm to 3 cm, not flush (confirmed by the data, Sec. 4.3); mock-up rotated around the fixed UMIK
Angle reference / temperature	Paper protractor under the mock-up; temperature not measured, $c = 344$ m/s assumed

Geometry as recorded in `LabB/data/labB_log.md`. The lab PC clock runs ahead of real time (files stamped 12:31–13:11 for a slot that ended at 12:00); the order of the runs is what matters and is preserved.



Setup: the mock-up is rotated about the fixed UMIK; $\theta = 0^\circ$ is the face towards the loudspeaker.

Mock-up dimensions were not taped; the brief's nominal values are used throughout.

3 Part 0: Anechoic chamber and free-field check

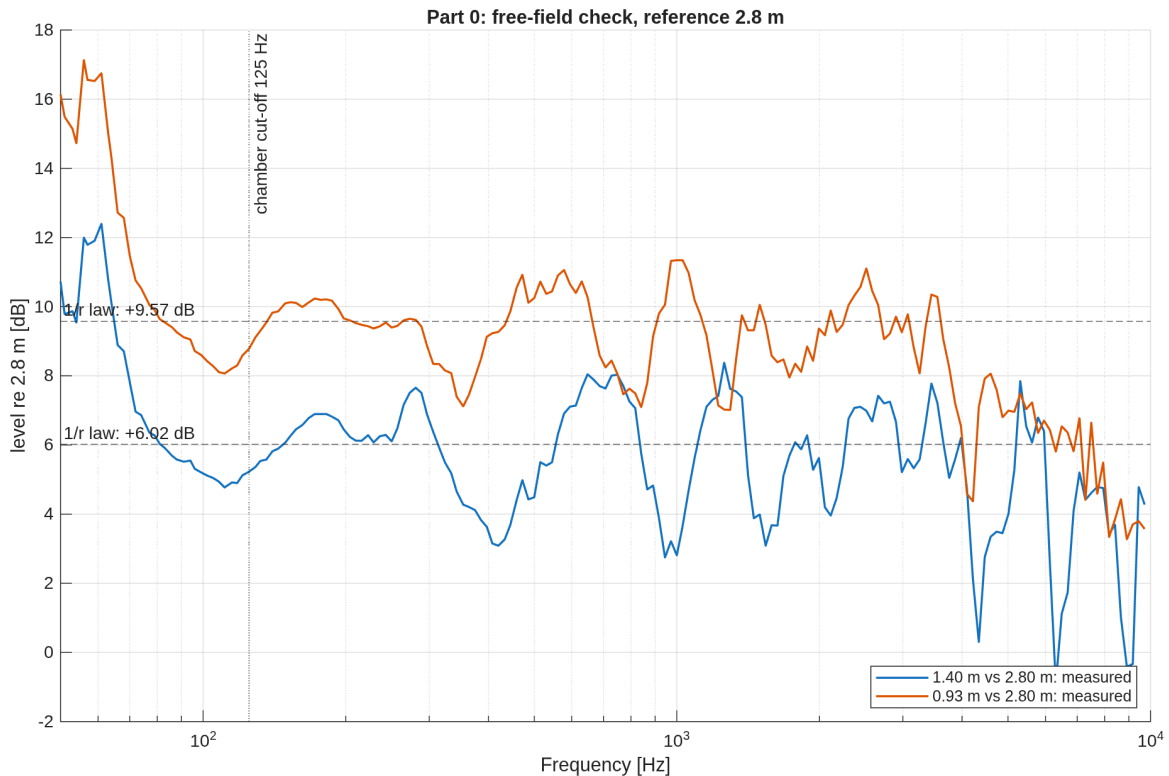
The brief specifies a chamber cutoff frequency of 125 Hz. The expected pressure magnitude of an approximately spherical wave varies as $1/r$, giving

$$\Delta L_{21} = 20 \log_{10} \left(\frac{|p(r_2)|}{|p(r_1)|} \right) \simeq 20 \log_{10} \left(\frac{r_1}{r_2} \right). \quad (4)$$

The distance check was carried out *after* the scattering series, without the mock-up: the UMIK was moved from its series position (2.8 m) to 1.4 m and 0.93 m. Doing it last guaranteed that the microphone did not move between the reference and the seven angles. The series position, 2.8 m, is taken as the reference, so the two closer positions should read $20 \log_{10}(2.8/r)$ higher.

Run	Distance / m	Data file	Level re 2.8 m / dB (mean 125 Hz to 200 Hz)
Reference distance	2.8	no_mockup_1.8m.mat → labB_dist280cm.mat	Reference
Second distance	1.4	no_mockup_1.4m.mat → labB_dist140cm.mat	+6.25 (predicted +6.02)
Third	0.93	no_mockup_0.93.mat → labB_dist93cm.mat	+9.84 (predicted +9.57)

The reference file is named `no_mockup_1.8m` on the lab PC; the tape reading was 2.8 m. The 0.93 m run is 9.8 dB above it. That matches 2.8 m (9.57 dB), not 1.8 m (5.7 dB), so the file name is a typo.



Part 0: level relative to the 2.8 m run (the series position), with the $1/r$ predictions (dashed).

How close to free field?

- **125–300 Hz:** both positions follow $1/r$ within -0.8 to $+1.6$ dB (1.4 m) and ± 0.8 dB (0.93 m). The band means are 0.2–0.3 dB above the prediction. Here the chamber behaves as a free field and the loudspeaker radiates like a point source.
- **300 Hz–4 kHz:** ripples of up to ± 3 dB around $1/r$, with a mean still within 0.6 dB. Much of the ripple is common to both curves, so it comes from the 2.8 m reference itself: interference between the direct sound and weak reflections from the stand, the floor net or the loudspeaker cabinet, which matter most at the largest distance, where the direct sound is weakest.

- **Above 4 kHz:** both closer positions read 2–4 dB below $1/r$ on average. At short range the microphone is no longer in the far field of the driver, and small offsets from the driver axis change the angle and hence the directivity. A 1 cm placement error is already 0.1λ .
- **Below the 125 Hz cut-off:** the closer positions read up to 6 dB to 8 dB above $1/r$. The wedges no longer absorb, and the room field adds to the direct sound, most strongly at the far position used as the reference.

For the scattering measurement this matters less than the ripple suggests. Every angle is normalised by the no-mock-up measurement at the *same* position, so a reflection that is common to both measurements cancels to first order. What remains is the error from reflections between the mock-up and the room, which the reference cannot remove.

4 Part 1: Scattering measurement procedure

1. The UMIK was connected before MATLAB was started; the course script found the internal card (`Line In, Speakers`) and the UMIK. A trial run (`test.mat`) was used to set the level.
2. The loudspeaker and the upward-pointing UMIK were fixed at 2.8 m; the mock-up was placed with its face centre 2 cm to 3 cm from the UMIK tip and rotated on a protractor for 0° , 30° , 60° , 90° , then 45° , 15° and 75° .
3. The mock-up was removed and the reference measured at the same UMIK position. No reference was taken *before* the series, so no drift check is available.
4. Part 0 (Sec. 3) was measured last. Data was copied to a group folder and verified at home.

The signal chain is the one in the sketch in Sec. 2.

Parameter	Value used	Parameter	Value used
f_1	50 Hz	f_2	10 000 Hz
Tones/octave	24 (184 tones)	Resolution	1 Hz
Averages N_{av}	32	UMIK incidence	90°

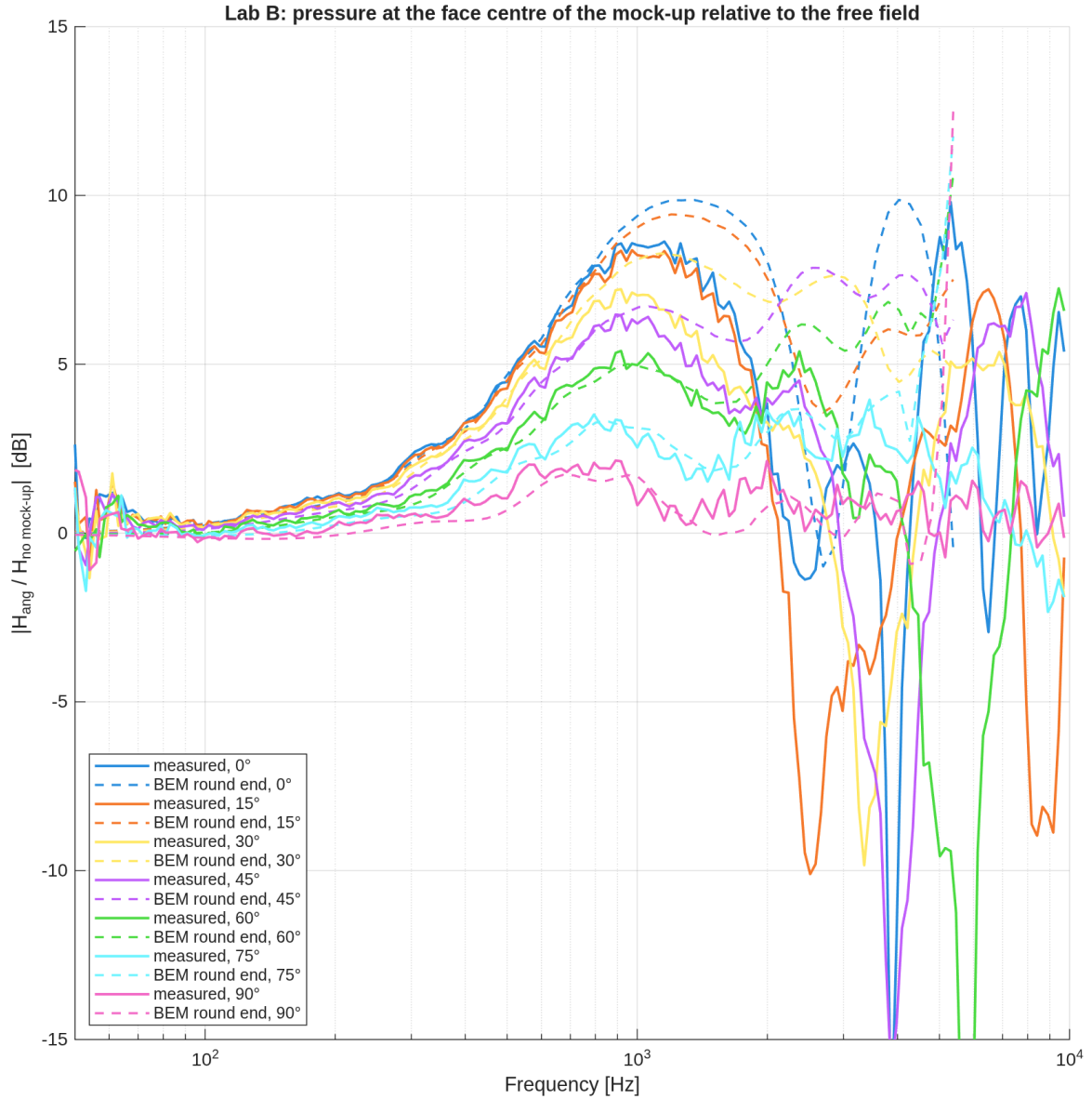
All as prescribed. The UMIK incidence selects its correction and is *not* the mock-up angle. Departure from the planned procedure: the course script `labB.m` was used instead of the group's wrapper, so each saved file holds only `h=specn(:,2)./specn(:,1)`; the frequency vector was reconstructed from the multitone generator with the same parameters (`import_group_files.m`), and the full spectra needed for a noise-floor plot were not saved.

4.1 Measurement record and quality checks

Condition	File (lab PC → analysed)	Angle	Quality / notes
Trial	test.mat	—	level check, not used
Mock-up	mockup_0.mat → labB_ang000.mat	0°	clean below 2 kHz; notch at 3.9 kHz (tip gap)
Mock-up	mockup_15.mat → labB_ang015.mat	15°	notch at 2.5 kHz
Mock-up	mockup_30.mat → labB_ang030.mat	30°	notch at 3.4 kHz
Mock-up	mockup_45.mat → labB_ang045.mat	45°	notch at 3.9 kHz
Mock-up	mockup_60.mat → labB_ang060.mat	60°	notch at 6.0 kHz
Mock-up	mockup_75.mat → labB_ang075.mat	75°	no deep notch
Mock-up	mockup_90.mat → labB_ang090.mat	90°	no deep notch
No mock-up	no_mockup_1.8m.mat → labB_nomockup.mat	—	reference, 2.8 m, taken after the series
End reference	not taken	—	no drift check possible

No repeats were needed. Clipping cannot be excluded from the saved data (no full spectra), but the ratios are smooth and consistent from angle to angle, and the trial run was used to set a moderate level.

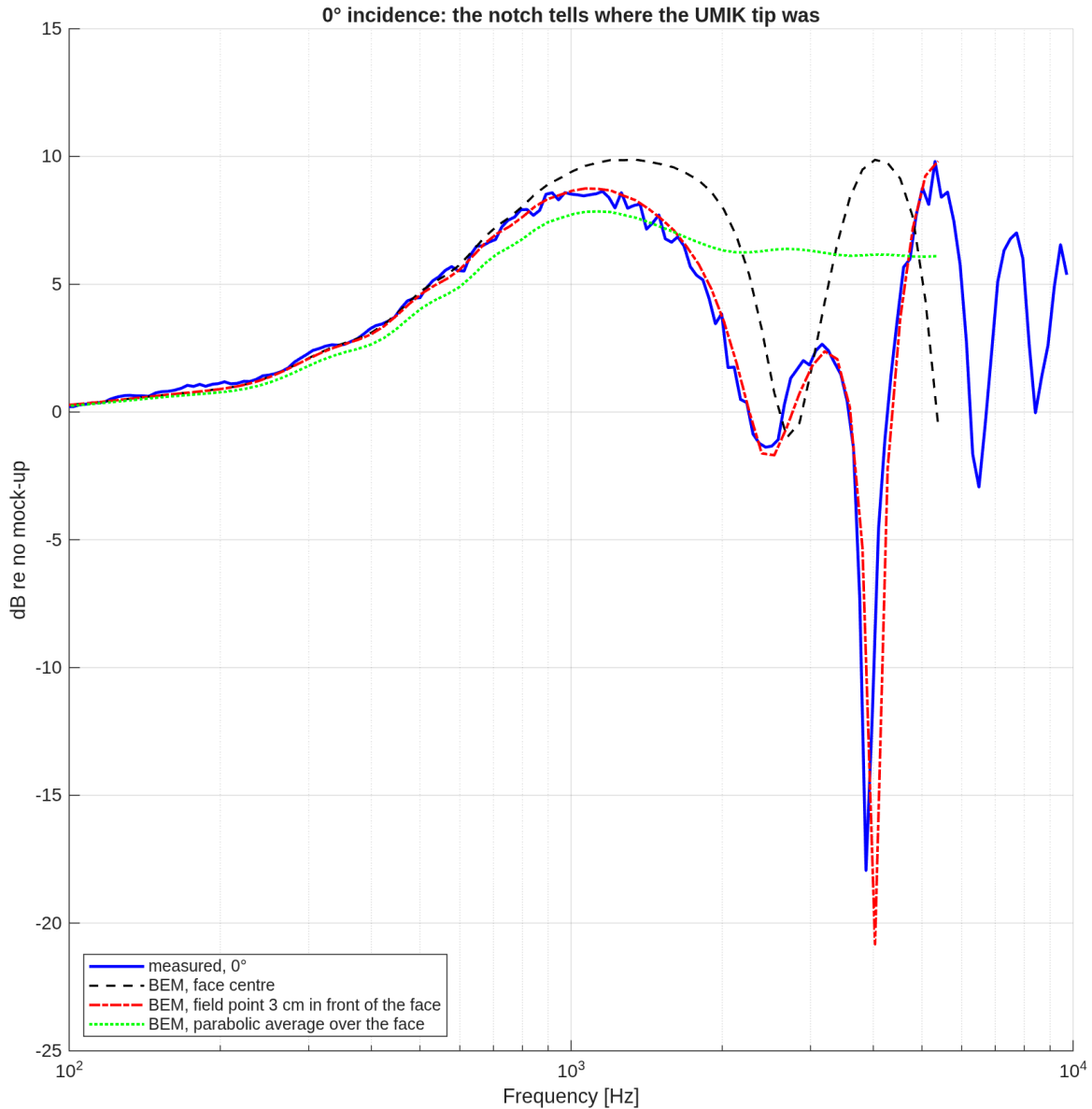
4.2 Normalized angle responses



Incidence	Level 125 Hz to 200 Hz / dB	Maximum below 2 kHz	At 1345 Hz: meas. / BEM	BEM face centre, max. below 2 kHz
0°	+0.8	+8.6 dB at 1155 Hz	+8.1 / +9.9	+9.9 dB at 1345 Hz
15°	+0.8	+8.4 dB at 972 Hz	+7.8 / +9.4	+9.4 dB at 1199 Hz
30°	+0.7	+7.2 dB at 917 Hz	+6.0 / +8.1	+8.3 dB at 1131 Hz
45°	+0.6	+6.5 dB at 891 Hz	+5.3 / +6.2	+6.7 dB at 1068 Hz
60°	+0.4	+5.4 dB at 917 Hz	+4.2 / +4.2	+5.0 dB at 951 Hz
75°	+0.2	+3.6 dB at 1995 Hz	+1.9 / +2.2	+3.3 dB at 848 Hz
90°	0.0	+2.1 dB at 891 Hz	+0.6 / +0.3	+1.7 dB at 673 Hz

Useful band: 125 Hz to 2000 Hz (ka from 0.3 to 4.6). Below 200 Hz every angle is within 1 dB of 0 dB: the body is acoustically invisible. From 200 Hz the response rises, head-on the fastest, and up to 2 kHz the measured curves follow the BEM for the same angles within about 1 dB and in the right order (0° highest, 90° lowest, within 2 dB). The 0° maximum is 1.3 dB lower and 15 % lower in frequency than the BEM centre-of-face value, all oblique curves fall away above 2 kHz, and there are notches of 10–19 dB between 2.5 and 6 kHz. These deviations are explained in the next section.

4.3 Position of the UMIK tip

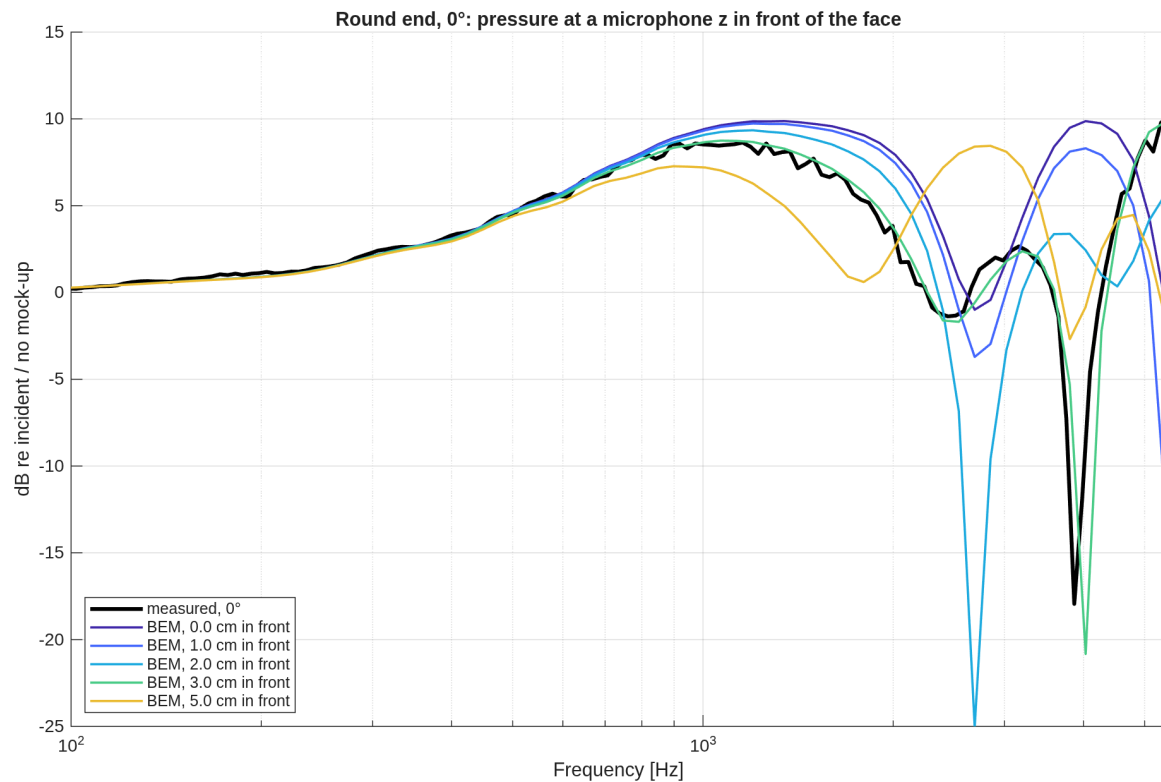


0°: measured (solid) against the BEM at the face centre, at a field point 3 cm in front of the face, and averaged over the face.

The BEM evaluated 3 cm in front of the face reproduces the measured 0° curve within 1.5 dB from 200 Hz to 3.5 kHz: the same maximum (+8.7 dB at 1.1–1.2 kHz), the same dip to –1.5 dB at 2.5 kHz, a notch of –21 dB at 4.0 kHz against the measured –18 dB at 3.9 kHz, and the return to +9 dB at 5 kHz. A rigid face reflects the wave onto itself and a standing wave forms in front of it; a quarter wavelength in front the incident and reflected waves cancel ($\lambda/4 = 2.2$ cm at 3.9 kHz), consistent with the recorded gap of 2–3 cm. At oblique incidence the path difference shrinks with $\cos\theta$ and the notch moves up in frequency; the field-point BEM places the 15°, 30° and 45° notches at 2.7, 3.4 and 4.3 kHz (measured 2.5, 3.35 and 3.9 kHz). The notch depths at oblique angles do not agree, but the BEM uses 8 circumferential terms and its oblique curves are not reliable above about 4 kHz. The measured data is therefore a centre-of-face free-field correction up to about 2 kHz and a measurement of the field 2 cm to 3 cm in front of the body above that.

4.4 How far in front of the face was the microphone? (BEM gap sweep)

The BEM field point was moved along the axis in front of the face centre (LabB/bem/explore_gap.m, round end, 0°) and each result compared with the measured 0° curve.



BEM pressure at 0, 1, 2, 3 and 5 cm in front of the face (colours) and the measured 0° curve (black).

Gap in front of the face	Maximum below 2 kHz	Deepest dip above 2 kHz
0 cm (face centre)	+9.9 dB at 1345 Hz	−1.0 dB at 2691 Hz
1 cm	+9.7 dB at 1199 Hz	−11.8 dB at 5382 Hz (grid edge)
2 cm	+9.3 dB at 1199 Hz	−25.0 dB at 2691 Hz
3 cm	+8.7 dB at 1068 Hz	−20.8 dB at 4032 Hz
5 cm	+7.3 dB at 898 Hz	−2.7 dB at 3805 Hz
Measured	+8.6 dB at 1155 Hz	−17.9 dB at 3868 Hz

Only the 3 cm curve has both the right maximum and a deep notch near 3.9 kHz; 2 cm puts the notch far too low and 5 cm lowers the maximum by more than 1 dB. The BEM grid is 1/12 octave, so the notch frequency (and hence the gap) is resolved to about 6%: the gap was close to 3 cm, which agrees with the 2 cm to 3 cm noted after the lab and with the quarter-wave estimate of 2.2 cm (the quarter-wave rule ignores the diffracted wave from the rim, so it only gives the order of magnitude).

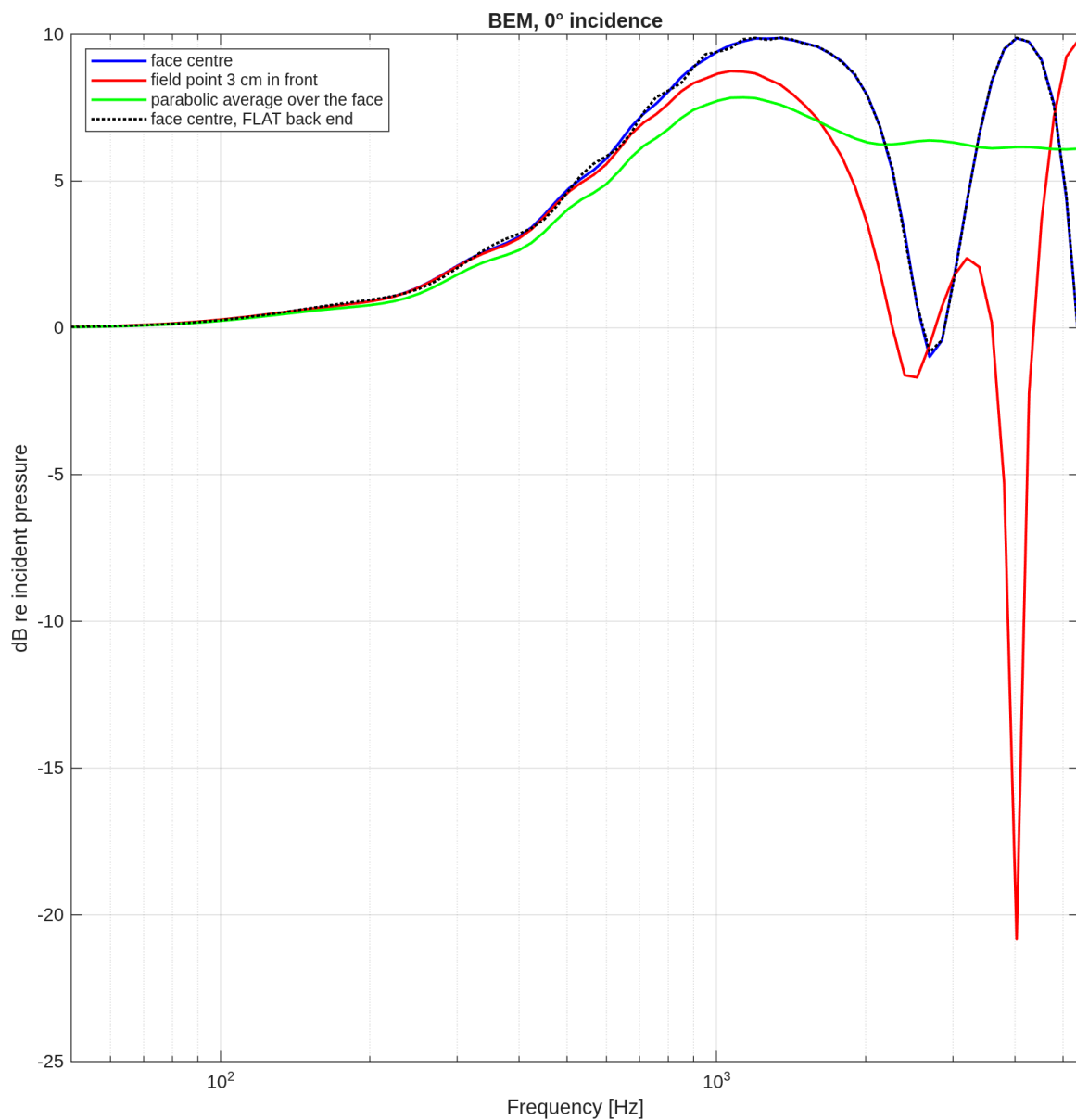
4.5 Reference stability (additional quality check)

No end reference was taken, so ΔL_{drift} cannot be evaluated. The indirect evidence is that every normalized curve sits at 0.0 to +0.8 dB between 125 and 200 Hz, in the order and at the level the BEM predicts, which a moved microphone or a changed gain would not give. Part 0 confirms that the gain did not change between the reference and the distance runs.

5 Part 2: BEM simulation and spatial effects

The supplied `CylinderPlaneWave` model assumes plane-wave incidence. Its frequency grid is 50 Hz to 5382 Hz in 1/12-octave steps. `AngleS` is in radians; `CylEnd` selects a round or flat back end. Geometry used: $D = 250$ mm, $L = 855$ mm, round back (`LabB/bem/run_bem.m`, angles 0° to 180°); the field point is 3 cm in front of the face centre on the axis. Comparison is made only over the common band 50 Hz to 5382 Hz.

Output / choice	Result at 0°
<code>p_centre</code>	+9.9 dB at 1345 Hz ($ka = 3.1$), dip to -1.0 dB at 2.7 kHz, back to +9.9 dB at 4 kHz.
<code>p_FP</code> (3 cm in front)	+8.7 dB at 1.1–1.2 kHz, -21 dB at 4.0 kHz: the measured curve.
<code>p_avg</code>	+7.8 dB maximum, +6.2 dB at 4 kHz: what a diaphragm of the full face diameter would feel; smoother at high ka .
Round vs. Flat	Less than 0.25 dB difference at 0° : the back end only matters for sound from behind.



BEM, 0° : centre, field point 3 cm in front, parabolic average over the face, and flat back end.

5.1 Agreement and limitations

Comparison	Observation / quantitative difference	Possible explanation
Measured vs. face centre	Within 1 dB up to 2 kHz at every angle; 0° maximum −1.3 dB and −15 % in frequency; notches above 2.5 kHz	Tip not flush (2–3 cm); protractor ± 2 –3°; spherical incidence; nominal diameter
Measured vs. field point	Within 1.5 dB up to 3.5 kHz at 0°, notch frequency within 4 %	This is the pressure that was actually measured
Centre vs. average	Average 2 dB lower at the maximum, 3.7 dB lower at 4 kHz	Pressure not uniform over the face at $ka > 3$; a real diaphragm averages
Round vs. flat rear	< 0.25 dB at 0°	Front face decides head-on

The field-point pressure best represents the experiment. Other contributions: the UMIK body and its stand scatter too (the ± 3 dB ripples above 400 Hz in Part 0 are already present without the mock-up and partly cancel in the ratio); the room below 125 Hz; the sound speed on the day (1 % per 3 °C on the frequency axis); and the numerical resolution of the BEM above 4 kHz at oblique angles.

6 Part 2 continued: Scale to real microphones

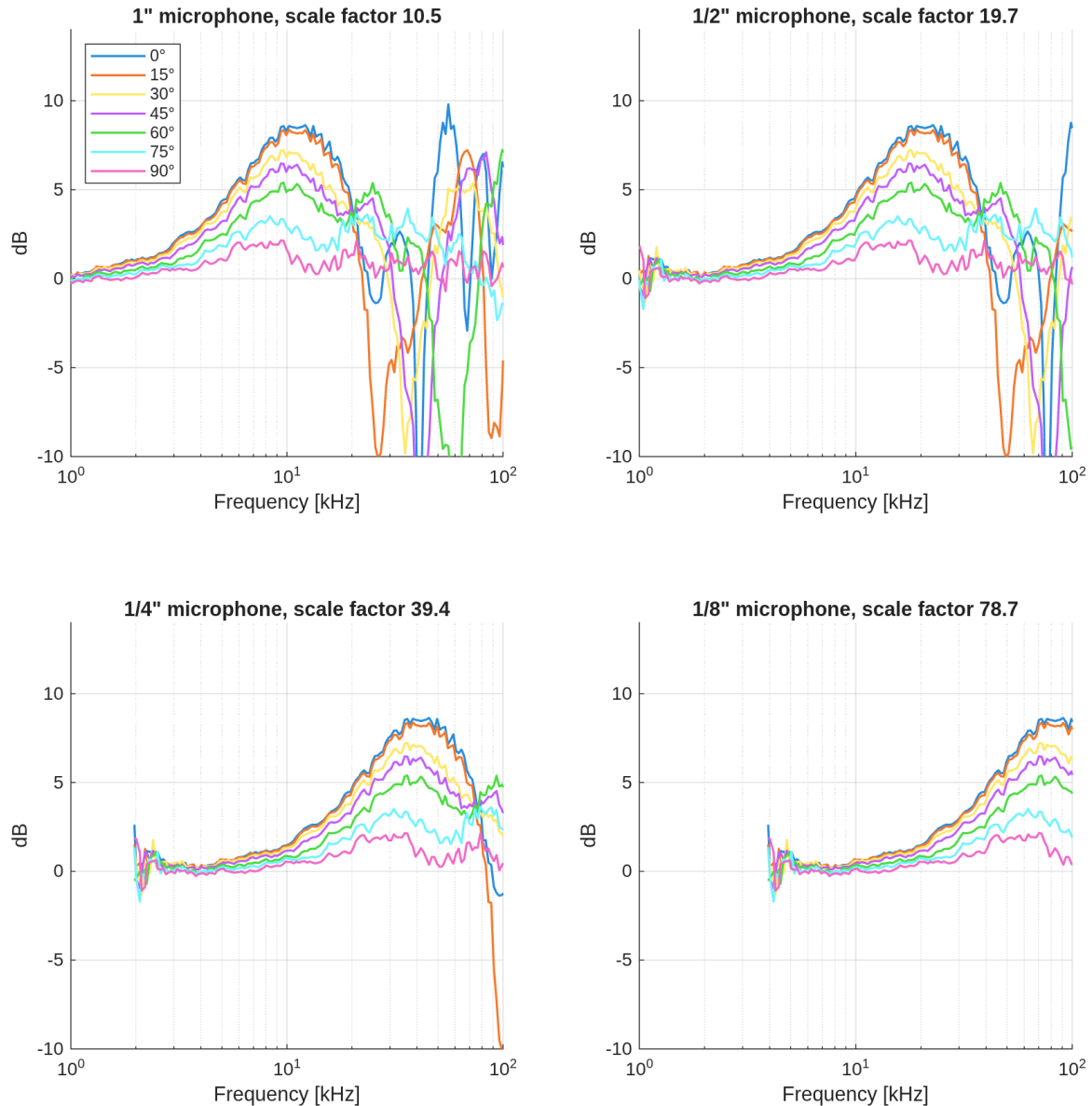
For a geometrically similar microphone of diameter D_{mic} , define

$$s = \frac{D_{\text{mock}}}{D_{\text{mic}}}, \quad f_{\text{mic}} = s f_{\text{mock}}. \quad (5)$$

The smaller microphone shows corresponding features at *higher* frequencies. Conversely, a microphone datasheet frequency maps to the mock-up axis as $f_{\text{mock}} = f_{\text{mic}}/s$. The normalized pressure level is unchanged under ideal geometric scaling with the same medium.

Nominal size	Diameter used / mm	Scale factor s	Measured 0° maximum lands at	Handbook type
1"	23.77	10.5	12.1 kHz	B&K 4144/4145
1/2"	12.70	19.7	22.7 kHz	B&K 4133/4134
1/4"	6.35	39.4	45 kHz	B&K 4135/4136
1/8"	3.175	78.7	91 kHz	B&K 4138

$D_{\text{mock}} = 250$ mm is the nominal value; the real microphone diameters are the nominal body diameters (the 1" body is 23.77 mm, not 25.4 mm).



The measured curves on the frequency axis of each microphone size.

The mock-up has no protection grid, so the comparison is with the handbook curves *without* grid (values read off the scanned figures in the brief, about ± 1 dB):

Microphone	Handbook figure (brief page)	Handbook 0° maximum	Mock-up scaled
1", B&K 4144/4145	Fig. 6.3, without grid (p. 6)	about +8 dB at 12–13 kHz; 90° within ± 1 dB to 10 kHz	+8.6 dB at 12.1 kHz; 90° +2.1 dB
1", B&K 4144/4145	Fig. 6.2, with grid (p. 6)	about +13 dB at 17 kHz	not comparable (grid)
1/2", B&K 4133/4134	Fig. 6.9, without grid (p. 9)	about +8.5 dB at 40–50 kHz	+8.6 dB at 22.7 kHz

For the 1" microphone without grid the scaled measurement lands on the handbook curve, both in level and frequency. The grid changes the curve completely (higher and sharper peak), which is why the no-grid figure is the right one. For the 1/2" microphones the level agrees but the handbook maximum sits higher in frequency than simple scaling predicts. Plausible reasons: a real 1/2" capsule is not a flat-faced cylinder (the diaphragm is recessed behind a rim), and the handbook value is what the whole diaphragm senses, which is closer to the face average p_{avg} than to the centre point, and the average rolls off more gently. The scaled BEM centre value would be +9.9 dB at 14 kHz for the 1" size.

7 Uncertainty and conclusion

Sources, ranked by their effect on the result:

1. UMIK tip 2 cm to 3 cm in front of the face: changes the result qualitatively above 2 kHz and lowers the 0° maximum by 1.3 dB. Systematic, explained by the BEM field point.
2. No start reference: drift cannot be quantified; the low-frequency levels (0.0 to +0.8 dB, in the BEM's order) bound it to below 1 dB; the budget below assumes ± 0.3 dB.
3. Mock-up diameter not measured: a few per cent on every frequency in the scaled comparison, nothing on the dB axis. For $s = D_{\text{mock}}/D_{\text{mic}}$,

$$\frac{u(s)}{s} = \sqrt{\left(\frac{u(D_{\text{mock}})}{D_{\text{mock}}}\right)^2 + \left(\frac{u(D_{\text{mic}})}{D_{\text{mic}}}\right)^2}. \quad (6)$$

4. Angle read from a paper protractor (± 2 – 3°): the curves are 0.5–1 dB apart per 15° near the maximum.
5. Spherical incidence from 2.8 m, temperature not recorded, room below 125 Hz: each below 1 dB or outside the useful band.

All normalized curves share one reference, so their errors are correlated: a reference error moves every angle together and does not change the angle dependence.

Numerical budget for the measured 0° maximum, $\Delta L = +8.6$ dB at 1155 Hz. No repeats were taken, so every entry is a Type B estimate; the *assumed* limits are stated so they can be challenged. The UMIK's own calibration cancels in the ratio (same microphone, same position for the reference and the angles). The tip gap is not in the budget: it is a known systematic effect, explained and quantified by the BEM (Sec. 4.3).

Contribution	Assumed limits, distribution	Sensitivity	u / dB
Drift between series and reference	± 0.3 dB (assumed from the 125–200 Hz levels), rectangular	1	0.17
Repeatability of one run	± 0.2 dB ripple on the smooth part of the curve, rectangular	1	0.12
Angle (paper protractor)	$\pm 3^\circ$, rectangular	0.013 dB/ $^\circ$ (0 to 15°)	0.02
Plane-wave approximation at 2.8 m	± 0.2 dB (assumed), rectangular	1	0.12
Combined standard uncertainty u_c			0.24
Expanded, $k = 2$ (about 95 %)			0.5

Result: $\Delta L_{0^\circ, \text{max}} = (8.6 \pm 0.5)$ dB ($k = 2$) at the UMIK position. On the frequency axis: the diameter (± 2 mm assumed, rectangular: 0.5 %) and the sound speed ($\pm 3^\circ\text{C}$: 0.5 %) combine to about 0.7 % standard

uncertainty in the scaled frequencies; the maximum itself is flat, so its frequency is only known to about $\pm 5\%$ from the curve shape.

Conclusion. Between 125 Hz and 2 kHz ($ka = 0.3$ to 4.6) the measured pressure increase at the face of the mock-up is 0 dB below $ka \approx 0.5$, rises head-on to +8.6 dB at 1155 Hz and stays within +2 dB at grazing incidence, in the order and within about 1 dB of the level the BEM predicts for the face centre. Above 2 kHz the measurement is that of a point 2–3 cm in front of the face, which the BEM field point reproduces within 1.5 dB up to 3.5 kHz including the 3.9 kHz notch. Scaled to a 1" microphone the maximum lands at 12 kHz, where the handbook correction has its +10 dB. Part 0 confirms free-field conditions to 0.1 dB between 125 and 400 Hz.

Sources and processing record

DTU 34870, *Lab B – Microphone scattering – E26*, 34870_Lab_B_CylinderScattering_E2026.pdf: pp. 1–5 (procedure, equations and BEM), followed by the microphone handbook excerpts. Source: course vault / DTU Learn.

Repository workflow: raw files from the lab PC in LabB/data/LabBmastergroup10/ (unchanged); LabB/matlab/import_group_files.m (frequency axis and metadata, writes labB_*.mat); LabB/bem/run_bem.m; LabB/matlab/process_labB.m with D_mockup=0.250 (nominal); LabB/bem/explore_gap.m for the gap sweep. Course measurement files: the 21 September 2026 re-upload (34870 - Lab B (2).zip).

Analysis version / selected repeats / deviations: processed 22 September 2026; no repeats; test.mat excluded; reference no_mockup_1.8m.mat identified as the 2.8 m run by the $1/r$ check; no start reference, hence no drift figure.